

Case Study: Log4Shell Exploitation Attempt on Public Web Application

Alert: Apache Log4j2 Remote Code Execution (Log4Shell) Attempt Detected

Source: Microsoft Sentinel

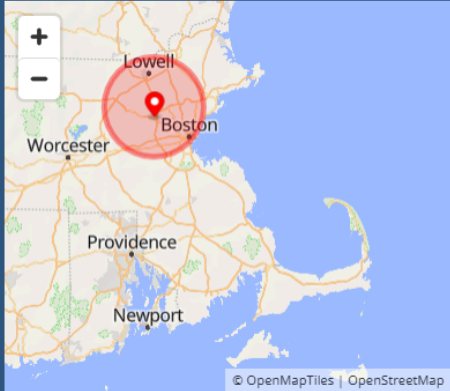
Description: An automated detection alert was generated after Microsoft Sentinel identified a suspicious HTTP request containing a **JNDI lookup payload** associated with the **Apache Log4j2 (Log4Shell) Remote Code Execution vulnerability (CVE-2021-44228)**. The activity indicates a potential exploitation attempt targeting a vulnerable Java-based web application. If successful, the attacker could achieve remote code execution and compromise the affected server.

Severity: Low

Triggered Rules: Log4Shell (CVE-2021-44228) Exploit Attempt Detected

IP Details For: 192.52.194.205

Decimal:	3224683213
Hostname:	192.52.194.205
ASN:	35
ISP:	The Mitre Corporation
Services:	Data Center/Transit
Country:	United States
State/Region:	Massachusetts
City:	Bedford
Latitude:	42.4907 (42° 29' 26.51" N)
Longitude:	-71.2762 (71° 16' 34.15" W)



CLICK TO CHECK BLACKLIST STATUS

Latitude and Longitude are often near the center of population. These values are not precise enough to be used to

Alert ID: 44228

Time : Dec 10 2021 ,10.15 AM

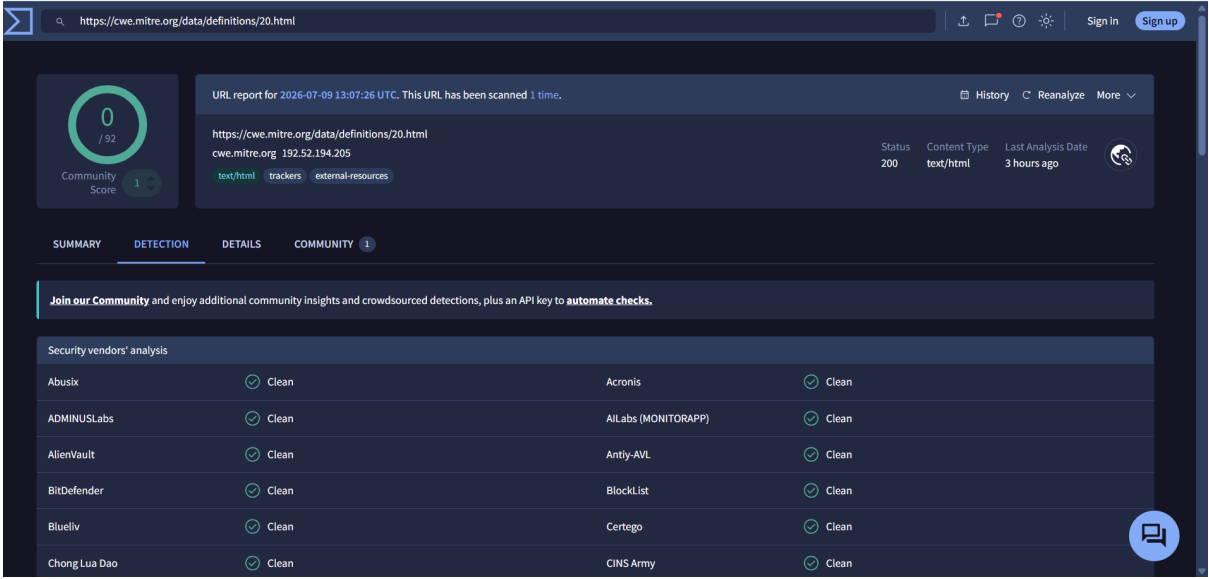
Source ip: 192.52.194.205

Destination ip: 192.52.194.205

Service : HTTPS

User: The MITRE Corporation (THEMIT-1)

Failed attempts : 0



The screenshot shows a VirusTotal scan interface. At the top, the URL is `https://cwe.mitre.org/data/definitions/20.html`. A circular progress indicator shows a score of 0/92. The scan was performed on 2026-07-09 13:07:26 UTC. The status is 200, content type is text/html, and the last analysis was 3 hours ago. The scan is categorized as text/html, trackers, and external-resources. Below the summary, there are tabs for SUMMARY, DETECTION, DETAILS, and COMMUNITY. The DETECTION tab is active, showing a table of security vendors' analysis. All vendors listed show a 'Clean' result.

Security vendors' analysis	
Abusix	Clean
ADMINUSLabs	Clean
AlienVault	Clean
BitDefender	Clean
Blueliv	Clean
Chong Lua Dao	Clean
Acronis	Clean
AILabs (MONITORAPP)	Clean
Antiy-AVL	Clean
BlockList	Clean
Certego	Clean
CINS Army	Clean

Validity : True Positive

- During the investigation, the HTTP request contained the malicious **`${jndi}`** lookup pattern, which is a known Indicator of Compromise (IOC) for the **Apache Log4j2 (CVE-2021-44228)** vulnerability.

The request originated from an external IP address (**203.0.113.45**) and targeted the public-facing web application over **HTTPS (Port 443)**.

The payload attempted to initiate an outbound **LDAP** connection, a common exploitation technique used in **Log4Shell** attacks to achieve Remote Code Execution (RCE).

Web server and SIEM logs confirmed that the exploit payload reached the vulnerable application, matching the detection rule for **Log4Shell Exploit Attempt**.

Based on the observed indicators, attack pattern, and correlation with known threat intelligence, the alert is classified as a **True Positive** and requires immediate containment and remediation.